

Custody Horizons for Ground-Based Angles-Only Tracking of Cislunar Periodic Orbits

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Draft, September 29, 2026

Abstract

Ground-based optical sensors are the most readily available assets for tracking objects in cislunar space, but they lose sight of those objects for days at a time. We quantify how long an object on a cislunar periodic orbit can go unobserved before it leaves a telescope search pattern, a quantity we call the *custody horizon* T_c . The study covers four representative orbits of the Earth–Moon system (the 9:2 southern near-rectilinear halo orbit, an L_1 northern halo, an L_2 planar Lyapunov orbit and a distant retrograde orbit) and a four-site network of 1 m-class telescopes over one year. Seen from Earth, these targets stay within $4\text{--}14^\circ$ of the Moon. As a result, the Moon-exclusion angle controls visibility more than telescope aperture does, and every orbit and network suffers a lunar-phase blackout of 8.5–16 days. Seven-day tracking arcs reach 1–17 km position uncertainty (Cramér–Rao bound), which EKF and UKF filters attain. After a 3.2-day gap, however, the EKF reacquires consistently in only 10 % of runs, compared with 100 % for a Gaussian-mixture UKF. Propagating post-fit covariances through gaps of up to 30 days with linear, unscented and Monte Carlo methods shows that custody lasts beyond 30 days for the NRHO and the DRO. The unstable L_1 halo and L_2 Lyapunov orbits lose custody after 11–17 days, depending mainly on how long the object was tracked before the gap: after 1-, 3- and 7-day arcs, 74–89 %, 96–100 % and 100 % of blackouts are survived with a ten-field search. The horizon is predicted almost exactly ($\log R^2 = 1.00$) by the unscented transform of the post-fit covariance, using 13 sigma points instead of Monte Carlo, but not by the orbit’s finite-time Lyapunov exponent alone ($R^2 < 0.05$). Linear (EKF-style) predictions give false custody: after 14 days on the L_1 halo their 99 % ellipse contains only 8–16 % of the true probability. These conclusions hold within 0.93–1.01 in a DE440 ephemeris model with solar radiation pressure, in which the orbits are transitioned by multiple shooting.

Keywords: cislunar space domain awareness; angles-only orbit determination; near-rectilinear halo orbit; uncertainty propagation; Gaussian mixture filter; custody

1 Introduction

Traffic beyond geosynchronous orbit is growing: lunar orbiters, the Gateway’s 9:2 near-rectilinear halo orbit (NRHO), and spacecraft in libration-point and distant retrograde orbits. Keeping track of these objects (*custody*) is harder than in near-Earth space for three reasons. The targets are faint and 380 000 km away. Seen from Earth, they sit close to the brightest object in the night sky. And their dynamics are governed by the Earth–Moon three-body problem, which stretches uncertainty rapidly and turns it non-Gaussian (Frueh et al., 2021).

Ground-based optical telescopes are the most accessible sensors for this task. Sechrest et al. (2024) detected CAPSTONE at $g \approx 19.6$ and Orion at $g \approx 15.5$ with 36 cm telescopes. But ground sensors see cislunar objects only intermittently: daylight, the Moon’s glare and the lunar phase open and close observation windows. Earlier work has shown that filters survive or fail

across particular gaps on particular trajectories (Chow et al., 2021; Iannamorelli and LeGrand, 2023). What is missing is a general answer to the operational question: *how long can a cislunar object go unobserved before it cannot be found again, and what determines that time?*

This paper answers that question for ground-based, angles-only tracking of periodic orbits. Its contributions are:

1. A physically generated observation model (sites, twilight, Moon exclusion and occultation, shadows, photometry with scattered moonlight). It shows that cislunar targets sit within 4–14° of the Moon and that a lunar-phase blackout of 8.5–16 days is unavoidable (Section 5).
2. A definition of the *custody horizon* T_c as the longest gap after which 99% of the true probability mass still lies inside a telescope search pattern. We give maps of T_c across orbits, tracking-arc lengths and search sizes (Section 7).
3. Evidence that T_c can be predicted without Monte Carlo from the stretching of the post-fit covariance (unscented transform, $\log R^2 = 1.00$), but not from the orbit’s finite-time Lyapunov exponent alone (Section 8).
4. A quantification of when the filter choice matters. EKF-style linear prediction gives false custody after about one week on unstable orbits, and a Gaussian-mixture UKF reacquires reliably where the EKF fails (Sections 6 and 8).
5. A cross-check in a DE440 ephemeris model with solar radiation pressure, using multiple-shooting ephemeris counterparts of the orbits (Section 9).

2 Related Work

Chow et al. (2021, 2022) processed ground-based angles-only and photometric observations from three sites for 15 periodic orbits. They found that Gaussian-mixture filters retain custody across data gaps where a UKF fails, and that two-body models lose the object after the first gap. Their gaps, however, were produced by decimating the data rather than by physical visibility. Frueh et al. (2021) showed that even a global ground network leaves periods with no visibility because of lunar brightness, and that cislunar uncertainty develops strongly curved, non-Gaussian structure. For space-based observers, Iannamorelli and LeGrand (2023, 2024) kept custody of a manoeuvring Artemis-like target across gaps of 1.7 and 2.6 days with an adaptive Gaussian-mixture interacting-multiple-model filter, in 247 of 250 Monte Carlo trials. For initial orbit determination, Jo et al. (2025) developed a three-body differential-correction method for angles-only data and reviewed the alternatives, including Gauss/Laplace extensions, particle-swarm, sparse-grid and admissible-region approaches. For observer placement, Wilmer and Bettinger (2022) studied NRHO surveillance from cislunar periodic orbits.

On the estimation side, we build on the unscented transform (Julier and Uhlmann, 2004; Wan and van der Merwe, 2000), entropy-triggered Gaussian-mixture splitting (DeMars et al., 2013), and regularised particle filtering with progressive correction (Oudjane and Musso, 2000; Musso et al., 2001). Consistency is assessed with the normalised estimation error squared (NEES) (Bar-Shalom et al., 2001). The dynamical tools are standard CR3BP methods (Szebeheley, 1967; Koon et al., 2011): Richardson’s third-order halo approximation (Richardson, 1980), differential correction and pseudo-arclength continuation. The finite-time Lyapunov exponent (FTLE) (Shadden et al., 2005) is the natural candidate for predicting how fast knowledge decays, and we test it directly.

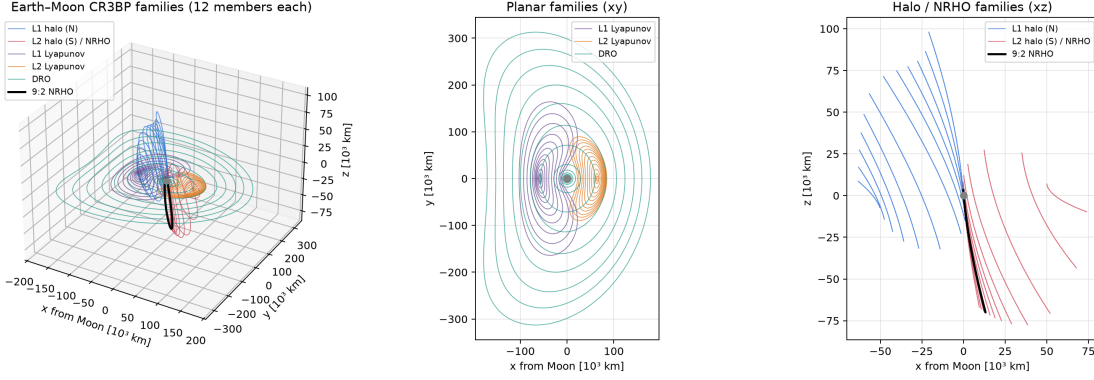


Figure 1: Earth–Moon CR3BP families used in this study (12 members each); the thick black curve is the 9:2 southern NRHO.

3 Models

3.1 Dynamics and orbit catalogue

Motion is modelled in the Earth–Moon circular restricted three-body problem (CR3BP), in the barycentric rotating frame with the Earth at $(-\mu, 0, 0)$ and the Moon at $(1 - \mu, 0, 0)$:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y}, \quad \ddot{z} = \frac{\partial \Omega}{\partial z}, \quad \Omega = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}, \quad (1)$$

with $\mu = 0.0121505842$ (from DE440 gravitational parameters), a length unit of 384 400 km and a time unit of 4.3425 days. The state transition matrix (STM) $\Phi(t, t_0)$ is integrated with the variational equations using an eighth-order Runge–Kutta method (Hairer et al., 1993; Virtanen et al., 2020). It agrees with Richardson-extrapolated finite differences to 3×10^{-10} .

Symmetric periodic orbits are computed by single-shooting differential correction and continued by pseudo-arclength continuation. Halo families are seeded with Richardson’s third-order solution and converge in four iterations. Five families are built: the L_1 northern halo (88 members), the L_2 southern halo down to the NRHOs (140), the L_1 and L_2 Lyapunov families (60 each) and the DRO family (94) (Fig. 1). The 9:2 synodic-resonant NRHO is found by period targeting. It has a period of 6.5624 d, perilune and apolune radii of 3249 and 71 222 km, and a stability index of 1.32, all consistent with Zimovan-Spreen et al. (2020). Four representative orbits are used throughout: the 9:2 NRHO, an L_1 halo with $A_z \approx 30\,000$ km, a mid-family L_2 Lyapunov orbit, and a DRO whose closest approach to the Moon is about 70 000 km.

3.2 Sky geometry

To give the sky geometry real dates, the CR3BP trajectory is placed in the instantaneous Earth–Moon frame of the actual Moon. The frame’s $\hat{x}$ axis points from Earth to Moon, its $\hat{z}$ axis lies along the Moon’s orbital angular momentum, and distances scale with the constant length unit. The Sun and Moon come from low-precision analytic series (Montenbruck and Gill, 2000), and Earth rotation from GMST. The Moon used in every geometric test is the mapped CR3BP Moon, so the target–Moon geometry is exactly the CR3BP one. Truth, filters and predictors share this mapping, so the setup is self-consistent. Section 9 replaces it with DE440 (Park et al., 2021). The reference epoch is 2027-01-01 UTC and the grid spacing is 10 min.

3.3 Sensors and visibility

The network consists of Hanle (India), La Palma (Spain), Haleakala (USA) and Siding Spring (Australia), in subsets of one site (India-1), three sites (Tri-3) and four sites (Tri-3+S). An

observation is possible only if all of the following hold: the Sun is below -18° ; the target is above 20° elevation; the line of sight misses the Moon; the target is outside Earth’s and the Moon’s (cylindrical) shadows; the target is more than θ_m from the Moon; and the target is brighter than the effective limiting magnitude. The target is a Lambertian sphere of radius 1 m and albedo 0.2 (magnitude 18.4 at zero phase at the Moon’s distance). The limiting magnitude is degraded by scattered moonlight following [Krisciunas and Schaefer \(1991\)](#), and extinction is applied. Dark-sky limiting magnitudes of 19.5, 21.5 and 23.0 are assumed for 0.36, 1 and 2 m apertures. The 0.36 m value is anchored to the CAPSTONE detection of [Sechrest et al. \(2024\)](#). Measurements are topocentric right ascension and declination with $1''$ noise.

3.4 Estimators

Four filters share one measurement model:

- an EKF, with STM covariance propagation and Joseph-form updates;
- a UKF (13 scaled sigma points, $\alpha = 1$, $\beta = 2$, $\kappa = 0$);
- a Gaussian-mixture UKF (GM-UKF). A component is split when its unscented prediction raises the differential entropy of its Gaussian fit by more than 0.02 nats ([DeMars et al., 2013](#)). The flow preserves volume, so any entropy gain comes from nonlinear bending. The split is a moment-preserving three-way split along the direction of maximum stretching (the top right-singular vector of the statistically linearised flow times the Cholesky factor of the covariance). Pruning and Mahalanobis merging keep the mixture small.
- a regularised bootstrap particle filter with progressive correction ([Oudjane and Musso, 2000](#); [Musso et al., 2001](#)). It hands over to a UKF once its position uncertainty falls below 50 km. Without the handover, a finite particle set impoverishes on long, dense arcs, and its covariance drops below the Cramér–Rao bound.

The truth has no process noise. The filters carry a small white acceleration noise with power spectral density $10^{-18} \text{ km}^2 \text{ s}^{-3}$.

4 Custody Horizon: Definition and Computation

Let $(\hat{\mathbf{x}}_0, P_0)$ be the post-fit estimate at the last measurement before a gap. For gap lengths $\Delta t \in [0.1, 30] \text{ d}$, the uncertainty is propagated three ways:

- *linear*: $P(\Delta t) = \Phi P_0 \Phi^\top$, which is what an EKF predicts;
- *unscented* (UT): 13 sigma points, which is what a UKF predicts;
- *Monte Carlo* (MC): samples of $\mathcal{N}(\hat{\mathbf{x}}_0, P_0)$ through the full dynamics, used as the reference.

The samples are projected onto the geocentric sky. With a search pattern of N square fields of view of side ϕ , whose equal-area radius is $r_N = \phi \sqrt{N/\pi}$, the custody horizon is

$$T_c(N) = \max \{ \Delta t : \Pr [\angle(\hat{\mathbf{u}}_{\text{target}}, \hat{\mathbf{c}}) \leq r_N] \geq 0.99 \}, \quad (2)$$

where $\hat{\mathbf{c}}$ is the centre of the search. Three variants use different centres or radii:

- *ideal*: the circle is centred on the true (MC) mean direction;
- *actual*: the circle is centred on a method’s own predicted direction;
- *claimed*: the time at which a method’s *own* 99% region outgrows r_N .

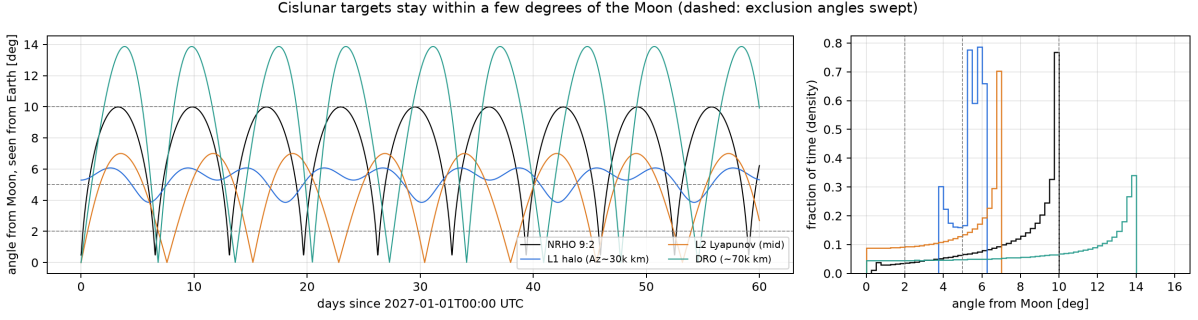


Figure 2: Geocentric angular separation between each target and the Moon: 60-day time series (left) and distribution over one year (right). Dashed lines mark the exclusion angles studied.

Table 1: One year of visibility (1 m telescopes, Tri-3+S network): share of time observable and longest observation gap, by Moon-exclusion angle.

Orbit	$\theta_m = 2^\circ$		5°		10°	
	visible	max gap	visible	max gap	visible	max gap
9:2 NRHO	37.5 %	12.8 d	31.4 %	13.2 d	4.5 %	19.5 d
L_1 halo	42.7 %	11.2 d	32.6 %	11.9 d	0 %	—
L_2 Lyapunov	32.7 %	13.5 d	20.6 %	14.5 d	0 %	—
DRO	38.0 %	11.8 d	32.5 %	11.9 d	21.2 %	16.5 d

A claimed horizon longer than the actual one is *false custody*: the predictor believes the target is still inside the pattern after it has left. We use $\phi = 1^\circ$ and $N \in \{1, 10, 100\}$, which give search radii of 0.56° , 1.78° and 5.64° . We also report Gaussian *containment*, the share of MC samples inside a method’s 99 % sky ellipse (ideal value 99 %).

The post-fit covariance at the gap start is the Cramér–Rao lower bound of the preceding tracking arc of 1, 3 or 7 days:

$$\mathcal{I}(t_0) = \sum_k \Phi(t_k, t_0)^\top H_k^\top R^{-1} H_k \Phi(t_k, t_0), \quad P_0 = \Phi(t_e, t_0) \mathcal{I}^{-1} \Phi(t_e, t_0)^\top, \quad (3)$$

built from the actual visible measurements, with the gap starting at the arc’s last measurement. Section 6 shows that the filters reach this bound. Two scenarios are studied: arcs ending every 15 days through the year (*phase*), and arcs ending exactly when a blackout of 2 days or more begins (*blackout*). This gives 525 cases.

We also test a purely dynamical predictor: $\hat{T}_{\text{FTLE}}$ is the first Δt at which the largest singular value of Φ , $s_{\max}(\Phi(\Delta t)) = e^{\sigma_{\Delta t} \Delta t}$, reaches r_N/θ_0 , where θ_0 is the initial 99 % sky radius and $\sigma_{\Delta t}$ is the FTLE (Shadden et al., 2005).

5 Visibility: the Moon and the Lunar Phase Dominate

Seen from Earth, the four targets stay close to the Moon (Fig. 2). The L_1 halo stays within 3.9 – 6.1° and the L_2 Lyapunov orbit within 0 – 7.0° . The NRHO ranges over 0.5 – 10.0° and spends most of its time near apolune, at about 10° . The DRO ranges over 0 – 13.9° . The Moon-exclusion angle is therefore the dominant sensor parameter (Table 1). With a 10° exclusion, the L_1 and L_2 targets are never observable, whereas at 5° they are observable 21–33 % of the year. Doubling the aperture from 1 to 2 m changes the visible fraction by only 0.3–1.6 percentage points.

Visibility arrives in windows that run from roughly first quarter to last quarter (Fig. 3). Near new Moon the target lies close to the Sun on the sky and is up only in daylight. Near full Moon the exclusion angle and the scattered moonlight bite. Gap lengths have three regimes (Fig. 4):

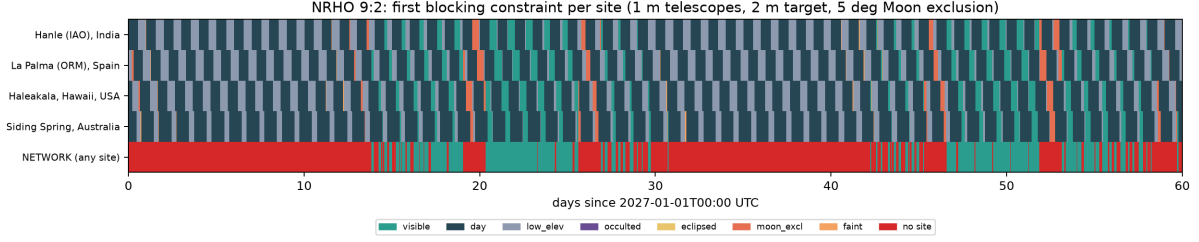


Figure 3: First blocking constraint per site for the 9:2 NRHO over 60 days (1 m, 5° exclusion). The bottom row shows network availability.

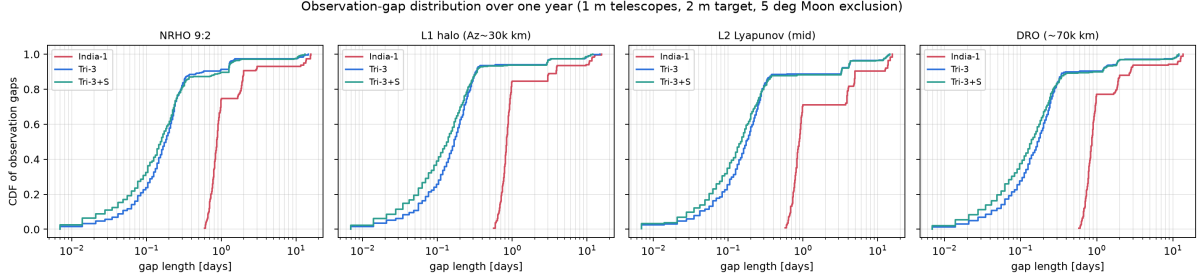


Figure 4: Distribution of observation gaps over one year (1 m, 5°) for the three networks.

handovers between sites (under 0.4 d), diurnal gaps of about 1 d, which dominate for a single site, and monthly blackouts of 3–15 d. Adding sites or aperture removes short gaps but never the monthly blackout. The longest gap is 8.5–16 d in every network–aperture–orbit combination at 5° .

6 Observability and Filter Performance

Cramér–Rao bounds. Figure 5 shows the bound, computed from the actual measurement schedule, for arcs starting every 2 days. At the end of 7-day arcs the median worst-direction position uncertainty is 1.12 km (NRHO), 1.74 km (L_1 halo), 6.87 km (DRO) and 17.1 km (L_2 Lyapunov). The corresponding velocity uncertainties are 6–100 mm s^{-1} . Between 75 % and 84 % of 7-day arcs fully determine the orbit. For the DRO and the Lyapunov orbit, the weakest direction lies within $7\text{--}8^\circ$ (median) of the line of sight, i.e. it is range. For the NRHO and the halo it is spread more widely (median $13\text{--}18^\circ$), because the strongly curved dynamics over a week also constrain range. Arcs that end inside a blackout show the uncertainty growth that motivates the rest of the paper.

Tracking. When the filters are started at the first measurement of the arc (from a 1000 km / 10 ms^{-1} prior), both the EKF and the UKF converge to the bound, as Table 2 shows for 20 Monte Carlo runs per orbit. The averaged NEES is close to the ideal value of 6. The NRHO’s value of 4.4–4.5 is slightly conservative, because the filters carry process noise that the truth lacks. The EKF begins to lose consistency only on the unstable orbits.

Reacquisition after a gap. In the L_1 halo arc the first measurement arrives 3.2 d after the prior epoch. By then the 1000 km prior has stretched into a curved cloud of about $40\,000 \times 12\,000$ km (Fig. 6). Only 10 % of EKF runs end consistent, with errors of up to 562 km while the filter claims about 2 km. The UKF reaches 70 %. The GM-UKF reaches 100 % (with a maximum error of 2.4 km), splitting into up to 25 components along the stretching direction and averaging 3.1 components over the arc. The particle filter with UKF handover reaches 70 %

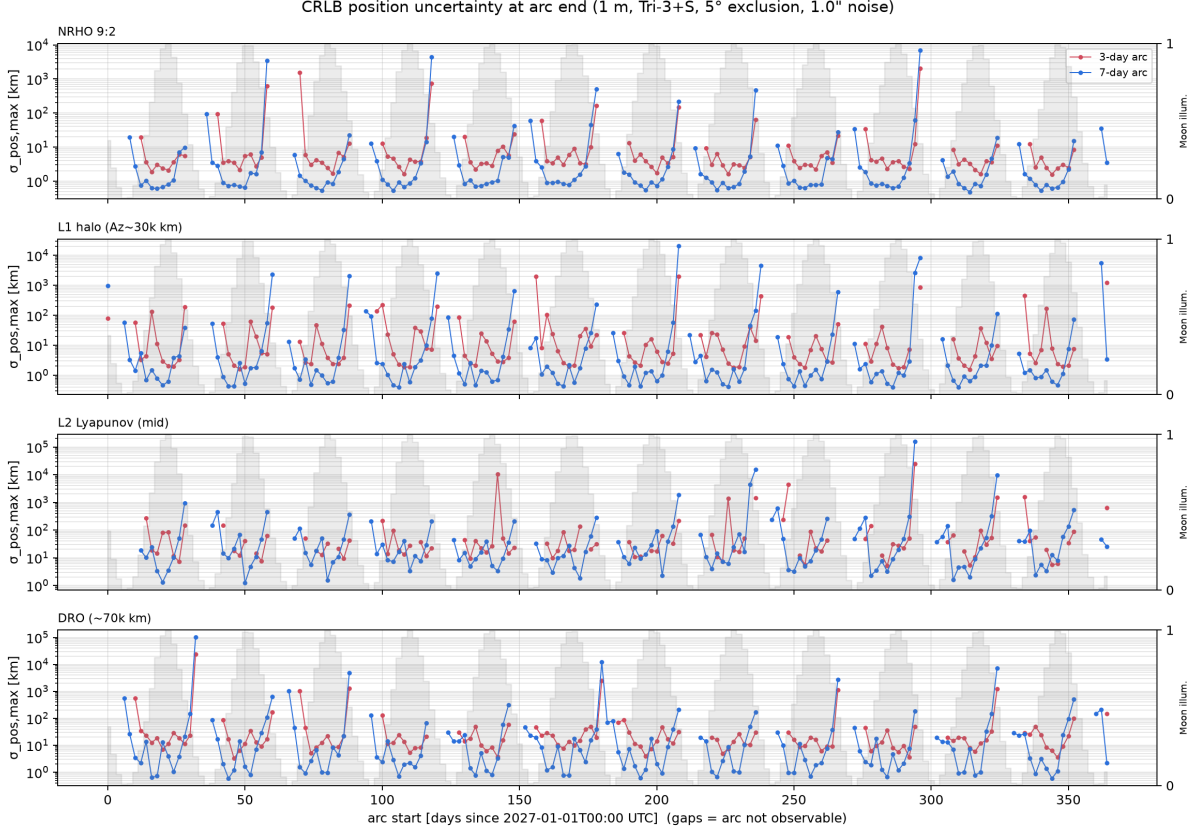


Figure 5: Cramér–Rao bound on the worst-direction position uncertainty at the end of 3-day and 7-day arcs, against the arc start date. Shading shows lunar illumination; gaps are arcs that are not observable.

(Fig. 7). The single-Gaussian 99 % ellipsoids still contain 96–98 % of the particles at the first measurement. At this gap length the EKF therefore fails mainly in the *measurement update*, which linearises the angles across a 10^4 km cloud, rather than in the prediction.

7 Custody Horizons

Figure 8 shows how the true 99 % sky radius grows after 7-day arcs. The NRHO and DRO stay below 0.01° for 30 days, reflecting their near-stability. The L_1 halo and L_2 Lyapunov orbits grow roughly exponentially after about 5 days and cross the search radii at 13–20 days. Because the growth is so steep, enlarging the search from 10 to 100 fields gains only 2–4 days (Table 3). The length of the tracking arc matters more: the L_1 halo’s median horizon rises from 10.9 to 16.5 d (10 fields) as the arc grows from 1 to 7 days.

Blackouts. The operational question is whether custody survives the blackouts that actually occur. Figure 9 and Table 4 compare T_c with every blackout of 2 days or more in the year. After 7-day arcs, every blackout is survived with a ten-field search. After 1-day arcs, 11–26 % of the blackouts are lost for the L_1 and L_2 targets, and 22 % for the NRHO, whose blackouts last about 10.5 d. The rule that follows is to track for at least 3 and preferably 7 days before a predicted blackout.

Table 2: Filter performance on 7-day arcs. Tracking: 20 runs, prior at the first measurement. Reacquisition (L_1 halo, 3.2d initial gap): 10 runs, prior at the arc start. “Consistent” means the final NEES is below $\chi^2_{0.99}(6)$.

<i>Tracking</i>					
Orbit	CRLB [km]	EKF error [km]	EKF ANEES	UKF error [km]	UKF ANEES
9:2 NRHO	1.13	1.10	4.51	1.08	4.40
L_1 halo	1.74	2.33	9.12	2.10	7.66
L_2 Lyapunov	17.60	15.25	6.35	15.15	6.31
DRO	6.95	6.88	6.23	6.93	6.25

<i>Reacquisition, L_1 halo</i>					
Filter	consistent	median error [km]	max error [km]	ANEES	time/run [s]
EKF	10 %	15.6	561.8	4.1×10^4	0.1
UKF	70 %	1.92	241.9	6.2×10^3	0.1
GM-UKF	100 %	1.14	2.4	7.5	0.2
PF→UKF	70 %	1.15	4.4	11.5	0.3

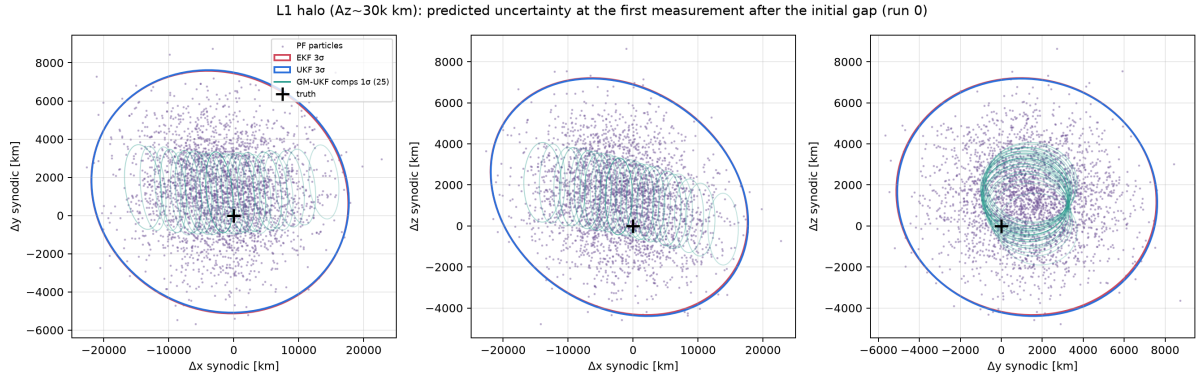


Figure 6: L_1 halo: predicted uncertainty at the first measurement after a 3.2-day gap. Particles, EKF/UKF 3σ ellipses and GM-UKF components (1σ) are shown relative to the truth.

8 Predicting the Custody Horizon, and False Custody

Can T_c be predicted without Monte Carlo? Figure 10 compares the Monte Carlo horizon with two cheap predictors. The *orbit-only* FTLE predictor fails, with $\log R^2$ between -0.22 and 0.03 . The *covariance-weighted* predictors succeed. The UT predictor reaches $R^2 = 0.98$, 1.00 and 0.86 for $N = 1$, 10 and 100 , and the linear predictor reaches 0.98 , 0.90 and 0.95 . The FTLE measures stretching along the flow’s most unstable direction, but the post-fit covariance is highly anisotropic: angles-only data constrain some directions far better than others, and only the uncertain directions that project onto the sky matter for the search. Predicting custody requires the growth of the *actual* covariance, which the unscented transform supplies from 13 sigma points instead of hundreds of samples.

Linear prediction is adequate for about a week but then breaks down on unstable orbits (Fig. 11). After 14 days, the L_1 halo’s linear 99 % sky ellipse contains only 7.6 % of the true probability mass (median), while the UT ellipse contains 98 %. A linear predictor used to plan the search would therefore point the telescopes at the wrong part of the sky. For all orbits the UT never produced false custody at $N = 10$. Beyond about 15 days, even the UT ellipse loses containment, falling to 70–80 %. The recovery near 30 days is an artefact of the small-angle projection once the distribution covers tens of degrees, long after custody is lost.

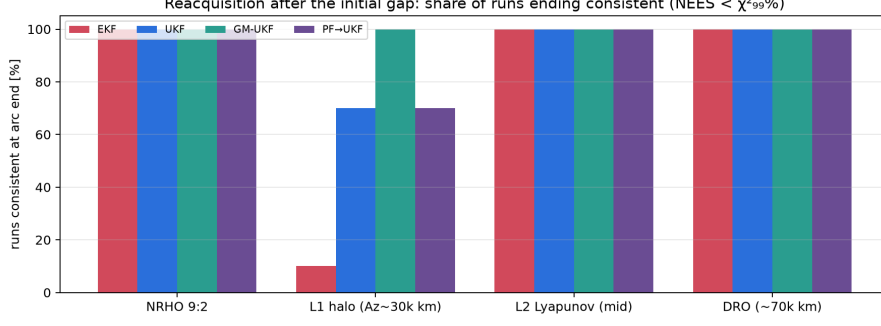


Figure 7: Share of reacquisition runs ending consistent, by filter and orbit.

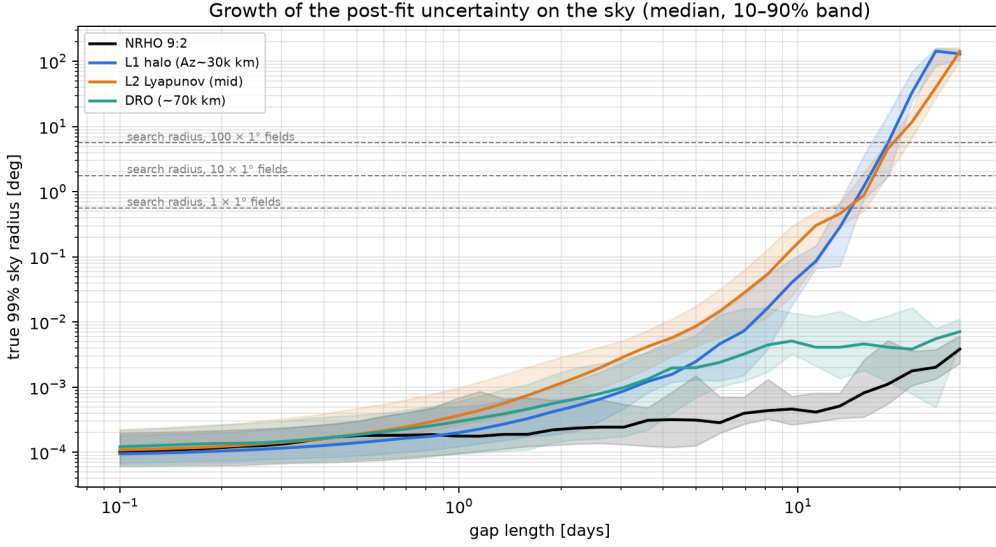


Figure 8: Growth of the true (Monte Carlo) 99 % sky radius after 7-day arcs: median and 10–90 % band over the year. Dashed lines are the search radii for 1, 10 and 100 one-degree fields.

9 Ephemeris Cross-Check

The CR3BP results are checked in an Earth-centred ephemeris model with DE440 Sun and Moon point masses (Park et al., 2021) and cannonball solar radiation pressure (area-to-mass ratio $0.01 \text{ m}^2 \text{ kg}^{-1}$, $C_r = 1.3$, cylindrical Earth shadow). Each orbit is first transitioned to its ephemeris counterpart by multiple shooting over 45-day windows. Patch points are placed every 0.1 periods and corrected by Levenberg–Marquardt followed by minimum-norm Newton iterations, with STMs from the gravity-gradient variational equations. Plain minimum-norm Newton diverged, or wandered by up to 10^5 km , across NRHO perilune passages. The L_1 halo, L_2 Lyapunov and DRO counterparts are continuous to better than 0.4 m and stay within 3500–5200 km (median) of their CR3BP shapes. The NRHO counterpart retains a continuity residual of 50–124 km, because the corrector stalls in the ill-conditioned perilune segments. We report this as a limitation. For 60 custody cases (blackout and phase scenarios, 1- and 7-day arcs), the CRLB covariance is mapped to inertial coordinates and propagated in the ephemeris model.

The conclusions carry over (Figs. 12 and 13, Table 5). The median ratio of ephemeris to CR3BP horizon is 0.93–1.01, and the growth curves overlap; 3 of 60 cases are outliers, where local differences in the ephemeris dynamics shift T_c by several days. Blackout survival is unchanged except for two single-case differences within four-case groups. The UT predictor again reaches $R^2 = 1.00$ for $N = 1$ and 10, but only 0.68 for $N = 100$: at a search radius of about 5.6° the distribution is too curved for any single Gaussian. Linear false custody becomes *worse*. At 14

Table 3: Custody horizon T_c [days], median (10th–90th percentile) over arcs through the year (phase scenario, ideal horizon). “>30” means beyond the 30-day grid.

Orbit	arc	σ_0 [km]	$N = 1$	$N = 10$	$N = 100$
9:2 NRHO	1 d	23.8	>30 (17.2–>30)	>30 (18.2–>30)	>30
	3, 7 d	3.9, 0.9	>30	>30	>30
L_1 halo	1 d	20.0	8.8 (5.5–10.0)	10.9 (7.3–11.6)	13.1 (9.6–13.7)
	3 d	5.1	11.7 (8.0–12.4)	13.2 (10.2–14.3)	15.1 (13.4–17.0)
	7 d	0.7	14.4 (13.1–15.2)	16.5 (14.6–18.4)	18.4 (16.6–19.7)
L_2 Lyapunov	1 d	30.2	9.5 (7.7–11.2)	12.4 (9.9–15.3)	16.0 (13.1–18.4)
	3 d	15.5	10.5 (9.1–13.4)	14.9 (11.2–16.3)	18.3 (15.6–19.0)
	7 d	7.0	13.8 (12.3–15.7)	16.7 (15.5–18.4)	19.2 (18.3–21.3)
DRO	1 d	31.8	>30 (7.8–>30)	>30	>30
	3, 7 d	11.7, 1.4	>30	>30	>30

Table 4: Blackouts (≥ 2 d) survived, i.e. ideal $T_c \geq$ blackout length, by tracking-arc length.

Orbit	arc	n	median length	$N = 1$	$N = 10$	$N = 100$	UT-centred, $N = 10$
9:2 NRHO	1 d	9	10.5 d	78 %	78 %	78 %	78 %
	3 d	11	10.5 d	100 %	100 %	100 %	100 %
	7 d	11	10.5 d	100 %	100 %	100 %	100 %
L_1 halo	1 d	27	3.1 d	67 %	89 %	93 %	89 %
	3 d	27	3.1 d	93 %	96 %	96 %	96 %
	7 d	28	3.1 d	100 %	100 %	100 %	100 %
L_2 Lyapunov	1 d	34	4.1 d	65 %	74 %	94 %	74 %
	3 d	34	4.1 d	79 %	97 %	97 %	97 %
	7 d	34	4.1 d	91 %	100 %	100 %	100 %
DRO	1 d	13	11.3 d	54 %	85 %	85 %	85 %
	3 d	13	11.3 d	100 %	100 %	100 %	100 %
	7 d	13	11.3 d	100 %	100 %	100 %	100 %

days the linear ellipse holds 15.5 % of the mass for the L_1 halo and 70.8 % for the L_2 Lyapunov orbit (98.9 % in the CR3BP), while the UT ellipse holds 97–99 %.

10 Discussion

Operational rules. Four rules follow from the results.

1. Sensor design for cislunar tracking from the ground should minimise the usable Moon-exclusion angle (baffling, scattered-light rejection) before increasing aperture.
2. The monthly lunar-phase blackout cannot be closed by adding ground sites. Custody through it depends on the post-fit covariance, so tasking should concentrate at least 3–7 days of tracking before each predicted blackout.
3. Search planning after a gap should use unscented (or Gaussian-mixture) prediction. Linear prediction is safe only for gaps shorter than about a week on unstable orbits.
4. Reacquisition after a long gap should use a Gaussian-mixture filter, which costs about twice as much as a UKF and was the only filter to reacquire every run.

Because the UT predictor tracks the Monte Carlo horizon with $R^2 = 1.00$ at $N \leq 10$, it can be evaluated on board a tasking scheduler for every catalogued object at negligible cost.

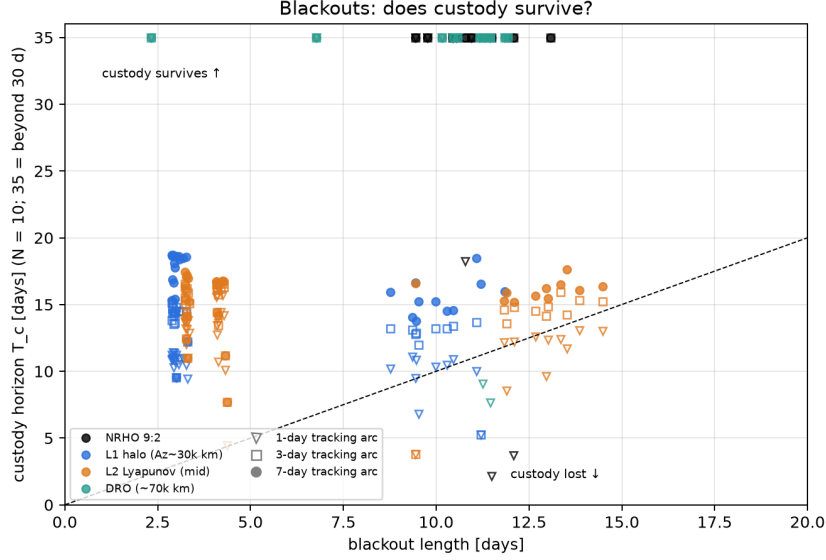


Figure 9: Custody horizon (10 fields) against blackout length. Points above the diagonal survive the blackout; marker shape shows the tracking-arc length.

Table 5: Ephemeris (DE440 + SRP) versus CR3BP: median custody horizon (10 fields) and median ratio, by tracking-arc length.

Orbit	arc	n	CR3BP T_c	ephemeris T_c	ratio
9:2 NRHO	1 d	8	>30 d	>30 d	0.98
	7 d	7	>30 d	>30 d	—
L_1 halo	1 d	4	8.5 d	7.8 d	0.93
	7 d	11	16.0 d	15.9 d	1.01
L_2 Lyapunov	1 d	6	12.4 d	11.3 d	0.95
	7 d	9	15.9 d	16.4 d	0.99
DRO	1 d	6	>30 d	>30 d	1.00
	7 d	9	>30 d	>30 d	—

Limitations.

- The post-fit covariance is the Cramér–Rao bound, which the filters reach in tracking but not always immediately after a gap.
- Sensor parameters (limiting magnitudes, extinction, dark-sky brightness) are nominal. The moonlight model of [Krisciunas and Schaefer \(1991\)](#) is extrapolated below 10° from the Moon, and the exclusion angle stands in for instrument-specific scattered light.
- Targets are passive, non-manoeuving 2 m spheres, and data association and clutter are not modelled.
- 99 % quantiles from 300 samples are noisy for individual cases, although the medians are stable.
- The NRHO ephemeris counterpart is not fully continuous.
- The CR3BP study places orbits in the real Earth–Moon direction with analytic ephemerides, and the DE440 check covers 60 of the 525 cases.

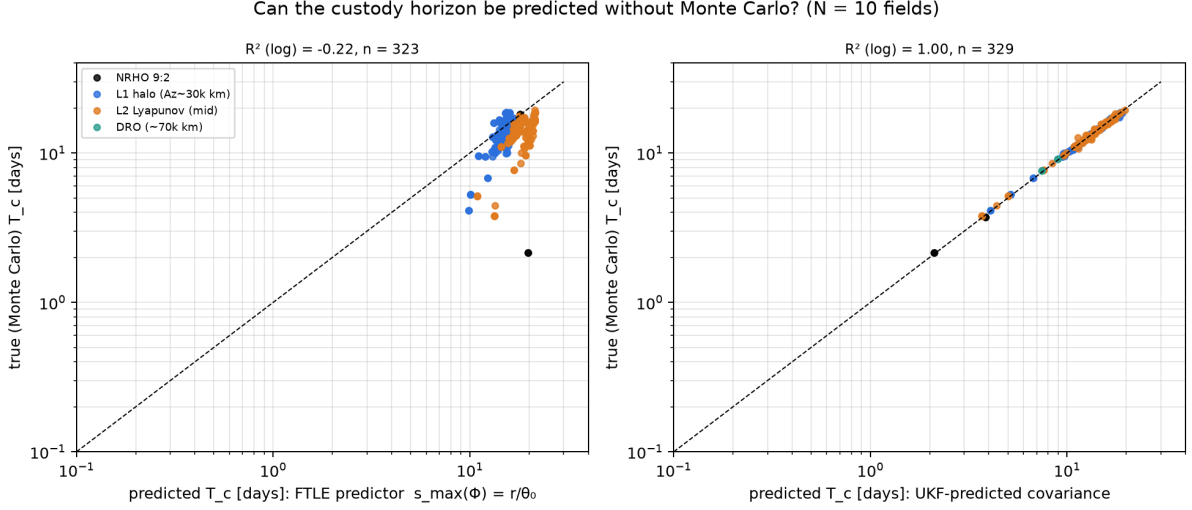


Figure 10: Monte Carlo custody horizon against the orbit-only FTLE predictor (left) and the UT-predicted covariance (right), for 10 fields.

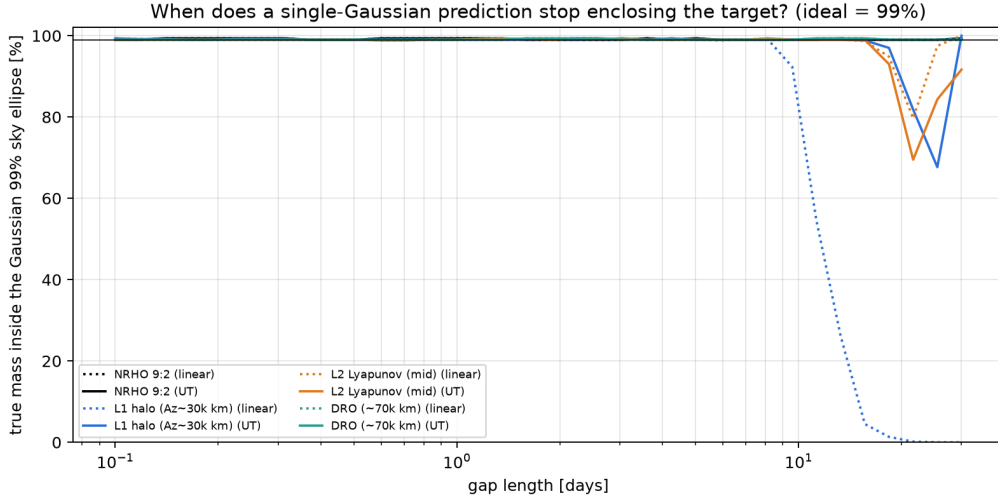


Figure 11: Share of the true probability mass inside the linear (dotted) and UT (solid) 99 % sky ellipses, against gap length (median over arcs). The ideal value is 99 %.

Future work. The main next step is validation with real observations. We plan to propose observing requests at Indian facilities (the 2 m HCT at Hanle, the ARIES telescopes at Devasthal and the GROWTH-India 0.7 m robotic telescope) for targets and nights selected from the custody maps. A Sun–Earth L_2 halo object such as JWST is a bright proxy target. Other extensions are custody-aware sensor tasking that uses the UT-predicted T_c , mixed ground and space-based networks, and manoeuvre detection.

11 Conclusions

We defined the custody horizon for ground-based, angles-only tracking of cislunar periodic orbits and mapped it over a year of physically generated observations. Cislunar targets sit within 4–14° of the Moon, so the exclusion angle and the lunar phase control visibility, and a blackout of 8.5–16 days is unavoidable. The near-stable 9:2 NRHO and DRO keep custody for more than 30 days after three or more days of tracking. The unstable L_1 halo and L_2 Lyapunov orbits keep it for 11–17 days (10 fields), with the tracking-arc length before a gap as the controlling factor.

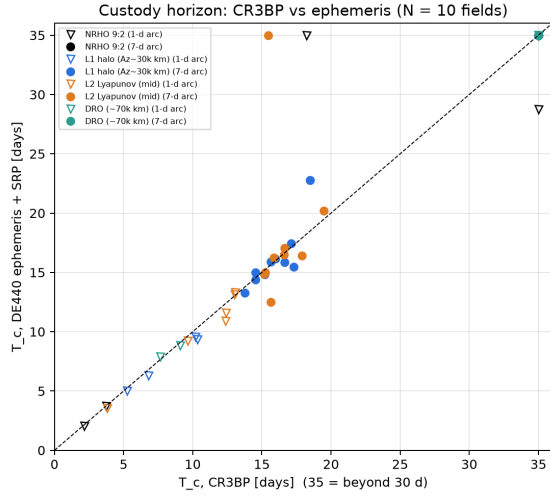


Figure 12: Custody horizon in the CR3BP and in the ephemeris model (10 fields).

The horizon is predicted almost exactly by the unscented transform of the post-fit covariance, but not by the orbit’s FTLE alone, and linear prediction gives false custody after about one week on unstable orbits. A Gaussian-mixture UKF reacquires reliably where the EKF fails. These conclusions hold in a DE440 ephemeris model with solar radiation pressure.

Code and Data Availability

The code (CR3BP and ephemeris dynamics, sensor model, estimators, custody analysis, 47 unit tests) and all scripts that regenerate the figures and tables are in the `cislunar-custody` repository, available from the author on request and to be released with publication.

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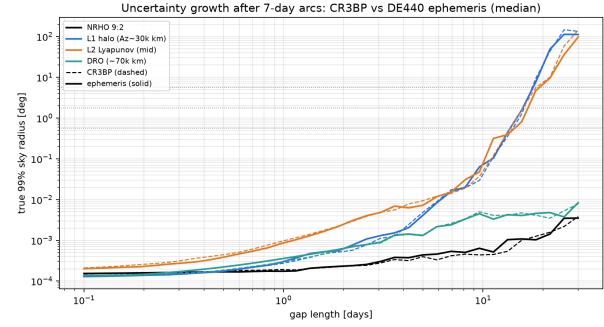


Figure 13: Median sky-radius growth after 7-day arcs: CR3BP (dashed) and ephemeris (solid).

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A Reference Parameters

Quantity	Value
CR3BP	$\mu = 0.0121505842$, length unit 384 400 km, time unit 4.3425 d
Epoch, grid	2027-01-01 00:00 UTC, 10 min, one year
Sites	Hanle (32.78°N, 78.96°E, 4.5 km); La Palma (28.76°N, 17.88°W, 2.4 km); Haleakala (20.71°N, 156.26°W, 3.05 km); Siding Spring (31.27°S, 149.06°E, 1.16 km)
Constraints	Sun $< -18^\circ$; elevation $> 20^\circ$; exclusion 2/5/10° (reference 5°); no occultation; sunlit
Telescopes	limiting magnitude 19.5 / 21.5 / 23.0 for 0.36 / 1 / 2 m (reference 1 m)
Target	Lambertian sphere, radius 1 m, albedo 0.2
Measurements	topocentric RA/Dec, $\sigma = 1''$, every 10 min per site when visible
Filters	prior 1000 km / 10 m s ⁻¹ ; process-noise PSD 10 ⁻¹⁸ km ² s ⁻³ ; UKF $\alpha = 1$, $\beta = 2$, $\kappa = 0$; GM split threshold 0.02 nats, ≤ 40 components; PF 3000 particles, hand-over at 50 km
Custody	fields of 1°, $N \in \{1, 10, 100\}$; 300 Monte Carlo samples; gaps of 0.1–30 d
Ephemeris	DE440 Sun and Moon, SRP $A/m = 0.01 \text{ m}^2 \text{ kg}^{-1}$, $C_r = 1.3$